

CS257 Linear and Convex Optimization

Homework 7

Due: November 9, 2020

November 2, 2020

1. Consider the quadratic function $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}^T \mathbf{A}\mathbf{x} - \mathbf{b}^T \mathbf{x}$, where $\mathbf{A} \succ \mathbf{O}$ is positive definite. Let $\mathbf{x}^*$ be the global minimum of f . Suppose that we can only compute the gradient up to some error. We implement the following gradient descent algorithm with constant step size t ,

$$\mathbf{x}_{k+1} = \mathbf{x}_k - t(\nabla f(\mathbf{x}_k) + \mathbf{e}_k),$$

where $\mathbf{e}_k \in \mathbb{R}^d$ is some additive error term (not to be confused with the k -th unit vector), satisfying a bound of the form $\|\mathbf{e}_k\|_2 \leq E$.

- (a). Let the eigenvalues of $\mathbf{A}$ be $\lambda_1 \geq \lambda_2 \geq \dots \lambda_n > 0$. If there is no error, i.e. $\mathbf{e}_k = \mathbf{0}$, what condition should t satisfy to ensure that $\mathbf{x}_k \rightarrow \mathbf{x}^*$?
- (b). Show $\mathbf{b} = \mathbf{A}\mathbf{x}^*$
- (c). Show $\mathbf{x}_{k+1} - \mathbf{x}^* = (\mathbf{I} - t\mathbf{A})(\mathbf{x}_k - \mathbf{x}^*) - t\mathbf{e}_k$
- (d). Assume $0 < t < 1/\lambda_1$ for (d)-(f). Show

$$\|\mathbf{x}_{k+1} - \mathbf{x}^*\|_2 \leq (1 - \lambda_n t) \cdot \|\mathbf{x}_k - \mathbf{x}^*\|_2 + tE.$$

Hint: Use $\|\mathbf{Q}\mathbf{x}\| \leq \lambda_{\max}(\mathbf{Q})\|\mathbf{x}\|$ for $\mathbf{Q} \succeq \mathbf{O}$ (see slide 14 of Lecture 8).

- (e). Show by induction that

$$\|\mathbf{x}_k - \mathbf{x}^*\|_2 \leq (1 - \lambda_n t)^k \|\mathbf{x}_0 - \mathbf{x}^*\|_2 + \frac{1 - (1 - \lambda_n t)^k}{\lambda_n} E$$

- (f). Conclude

$$\limsup_{k \rightarrow \infty} \|\mathbf{x}_k - \mathbf{x}^*\|_2 \leq \frac{E}{\lambda_n}$$

i.e. $\mathbf{x}_k$ “converges” to a ball of radius E/λ_n centered at $\mathbf{x}^*$.

Proof. (a). $0 < t < 2/\lambda_1$.

- (b). The first-order optimality condition is

$$\nabla f(\mathbf{x}^*) = \mathbf{A}\mathbf{x}^* - \mathbf{b} = \mathbf{0} \implies \mathbf{b} = \mathbf{A}\mathbf{x}^*$$

(c).

$$\nabla f(\mathbf{x}_k) = \mathbf{A}\mathbf{x}_k - \mathbf{b} = \mathbf{A}\mathbf{x}_k - \mathbf{A}\mathbf{x}^*$$

The update rule becomes

$$\mathbf{x}_{k+1} = \mathbf{x}_k - t(\mathbf{A}\mathbf{x}_k - \mathbf{A}\mathbf{x}^* + \mathbf{e}_k)$$

so

$$\mathbf{x}_{k+1} - \mathbf{x}^* = (\mathbf{I} - t\mathbf{A})(\mathbf{x}_k - \mathbf{x}^*) - t\mathbf{e}_k$$

(d). The eigenvalues of $\mathbf{I} - t\mathbf{A}$ are $1 - \lambda_1 t \leq 1 - \lambda_2 t \leq \dots \leq 1 - \lambda_n t$. Taking the norm of the equation in (c),

$$\begin{aligned} \|\mathbf{x}_{k+1} - \mathbf{x}^*\|_2 &= \|(\mathbf{I} - t\mathbf{A})(\mathbf{x}_k - \mathbf{x}^*) - t\mathbf{e}_k\|_2 \\ &\leq \|(\mathbf{I} - t\mathbf{A})(\mathbf{x}_k - \mathbf{x}^*)\|_2 + t\|\mathbf{e}_k\|_2 \\ &\leq (1 - \lambda_n t)\|\mathbf{x}_k - \mathbf{x}^*\|_2 + tE \end{aligned}$$

(e). Trivially true for $k = 0$. Now assume the inequality holds for k . For $k + 1$,

$$\begin{aligned} \|\mathbf{x}_{k+1} - \mathbf{x}^*\|_2 &\leq (1 - \lambda_n t)\|\mathbf{x}_k - \mathbf{x}^*\|_2 + tE \\ &\leq (1 - \lambda_n t) \left[(1 - \lambda_n t)^k \|\mathbf{x}_0 - \mathbf{x}^*\|_2 + \frac{1 - (1 - \lambda_n t)^k}{\lambda_n} E \right] + tE \\ &= (1 - \lambda_n t)^{k+1} \|\mathbf{x}_0 - \mathbf{x}^*\|_2 + \frac{1 - (1 - \lambda_n t)^{k+1}}{\lambda_n} E \end{aligned}$$

which completes the induction.

(f).

$$\limsup_{k \rightarrow \infty} \|\mathbf{x}_k - \mathbf{x}^*\|_2 \leq \frac{E}{\lambda_n}$$

□

2. Suppose $f(\mathbf{x})$ is differentiable and α -strongly convex, and $g(\mathbf{x})$ is β -smooth. Prove that the function $h(\mathbf{x}) = f(\mathbf{x}) - g(\mathbf{x})$ is convex if $\alpha \geq \beta$.

Proof. Since f is α -strongly convex,

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) + \frac{\alpha}{2}\|\mathbf{x} - \mathbf{y}\|^2$$

Since g is β -smooth,

$$g(\mathbf{y}) \leq g(\mathbf{x}) + \nabla g(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) + \frac{\beta}{2}\|\mathbf{x} - \mathbf{y}\|^2$$

Taking the difference of the above two inequalities, we obtain

$$h(\mathbf{y}) \geq h(\mathbf{x}) + \nabla h(\mathbf{x})^T(\mathbf{y} - \mathbf{x}) + \frac{\alpha - \beta}{2}\|\mathbf{x} - \mathbf{y}\|^2$$

If $\alpha \geq \beta$, this implies

$$h(\mathbf{y}) \geq h(\mathbf{x}) + \nabla h(\mathbf{x})^T(\mathbf{y} - \mathbf{x})$$

so h is convex. □

3. Consider $f(\mathbf{x}) = \frac{1}{2}\mathbf{x}_1^2 + (1 - \epsilon)\mathbf{x}_1\mathbf{x}_2 + \frac{1}{2}\mathbf{x}_2^2$, where $\epsilon \in (0, 1)$. Find the condition number of the Hessian of f . What happens to the condition number as $\epsilon \rightarrow 0$?

Proof. The Hessian matrix is

$$\nabla^2 f(\mathbf{x}) = \begin{pmatrix} 1 & 1 - \epsilon \\ 1 - \epsilon & 1 \end{pmatrix}$$

The eigenvalues are ϵ and $2 - \epsilon$. The condition number is

$$\kappa(\nabla^2 f(\mathbf{x})) = \frac{2 - \epsilon}{\epsilon}$$

As $\epsilon \rightarrow 0$, $\kappa(\nabla^2 f(\mathbf{x})) \rightarrow +\infty$. □